

2520090071

D.Vignesh Kumar

WEEK-4

BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

```
1 • CREATE DATABASE bookflow_db;
2 • USE bookflow_db;
3 • SELECT DATABASE();
4
5 • CREATE TABLE Books(
6     book_id INT AUTO_INCREMENT PRIMARY KEY,
7     title VARCHAR(200) NOT NULL,
8     author VARCHAR(150) NOT NULL,
9     genre VARCHAR(100),
10    description TEXT,
11    published_year INT
12 );
13
14 • DESCRIBE Books;
15
16
17
18 • INSERT INTO Books
19 (
20     title,
21     author,
22     genre,
23     description,
24     published_year
25 )
26 VALUES
27
```

100% 19:3

Action Output

	Time	Action	Response
✓ 1	12:36:08	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 2	12:36:11	USE bookflow_db	0 row(s) affected

PART 2 — Create the Books table

```
3 SELECT DATABASE();
4
5 CREATE TABLE Books(
6     book_id INT AUTO_INCREMENT PRIMARY KEY,
7     title VARCHAR(200) NOT NULL,
8     author VARCHAR(150) NOT NULL,
9     genre VARCHAR(100),
10    description TEXT,
11    published_year INT
12 );
13
14 DESCRIBE Books;
15
```

100% 19:3

Result Grid Filter Rows: Search Export:

DATA...
book...

Result 1

Action Output

	Time	Action
✓ 1	12:36:08	CREATE DATABASE bookflow_db
✓ 2	12:36:11	USE bookflow_db
✓ 3	12:36:35	SELECT DATABASE() LIMIT 0, 1000

```

5  CREATE TABLE Books(
6      book_id INT AUTO_INCREMENT PRIMARY KEY,
7      title VARCHAR(200) NOT NULL,
8      author VARCHAR(150) NOT NULL,
9      genre VARCHAR(100),
10     description TEXT,
11     published_year INT
12 );

```

```

14 DESCRIBE Books;

```

```

18 INSERT INTO Books
19 (
20     title,
21     author,
22     genre,
23     description,
24     published_year
25 )
26 VALUES

```

100% 3:12

Action Output

	Time	Action
✓ 1	12:36:08	CREATE DATABASE bookflow_db
✓ 2	12:36:11	USE bookflow_db
✓ 3	12:36:35	SELECT DATABASE() LIMIT 0, 1000
✓ 4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY,

14

15

16

100%

16:14

DESCRIBE Books;

Result Grid

Filter Rows:

Export:

	Field	Type	Null	Key	Default	Extra	
	book_id	int	NO	PRI	NULL	auto_increment	
	title	varchar(200)	NO		NULL		
	author	varchar(150)	NO		NULL		
	genre	varchar(100)	YES		NULL		
	description	text	YES		NULL		
	published_year	int	YES		NULL		

Result 2

Action Output

		Time	Action	Response
✓	1	12:36:08	CREATE DATABASE bookflow_db	1 row(s) affected
✓	2	12:36:11	USE bookflow_db	0 row(s) affected
✓	3	12:36:35	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓	4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY, title VAR...	0 row(s) affected
✓	5	12:37:54	DESCRIBE Books	6 row(s) returned

PART 3 — Insert the books

```
'A fantasy adventure involving Bilbo Baggins and a dangerous journey.',
1937
),
(
    'Enders Game',
    'Orson Scott Card',
    'Science Fiction',
    'A young student is trained for an interstellar conflict.',
    1985
),
(
    'Harry Potter and the Sorcerers Stone',
    'J.K. Rowling',
    'Fantasy',
    'A young wizard begins his magical education.',
    1997
),
(
    'Twenty Thousand Leagues Under the Seas',
    'Jules Verne',
    'Adventure',
    'A fantastic underwater adventure aboard a mysterious submarine.',
    1870
);
```

100% 1:107

Action Output

	Time	Action	Response
✓ 1	12:36:08	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 2	12:36:11	USE bookflow_db	0 row(s) affected
✓ 3	12:36:35	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY, title VAR...	0 row(s) affected
✓ 5	12:37:54	DESCRIBE Books	6 row(s) returned
✓ 6	12:38:33	INSERT INTO Books (title, author, genre, description, published_year) V...	10 row(s) affected Record

108

SELECT * FROM Books;

100%

21:108

Result Grid

Filter Rows:

Edit:

Export/Import:

	book_id	title	author	genre	description	published_ye...
	1	The Martian	Andy Weir	Science Fiction	An astronaut struggles to survive alone on Mars.	2011
	2	Dune	Frank Herbert	Science Fiction	A story involving planets, politics, survival and s...	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	A humorous adventure across space and alien...	1979
	4	Pride and Prejudice	Jane Austen	Romance	A classic story about relationships and social ex...	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	A story of ambition, wealth, love and the Americ...	1925
	6	Foundation	Isaac Asimov	Science Fiction	A futuristic story about mathematics, civilization...	1951
	7	The Hobbit	J.R.R. Tolkien	Fantasy	A fantasy adventure involving Bilbo Baggins and...	1937
	8	Enders Game	Orson Scott Card	Science Fiction	A young student is trained for an interstellar con...	1985
	9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	A young wizard begins his magical education.	1997
	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	A fantastic underwater adventure aboard a myst...	1870

Books 3

Action Output

	Time	Action	Response
✓ 2	12:36:11	USE bookflow_db	0 row(s) affected
✓ 3	12:36:35	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY, title VAR...	0 row(s) affected
✓ 5	12:37:54	DESCRIBE Books	6 row(s) returned
✓ 6	12:38:33	INSERT INTO Books (title, author, genre, description, published_year) V...	10 row(s) affected Records: 10 Duplicates: 0 V
✓ 7	12:39:51	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned

PART 4 — Add the embedding column

120ALTER TABLE Books

121ADD COLUMN embedding JSON;

122

100%1:120

Action Output

	Time	Action	Response
3	12:36:35	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY, title VAR...	0 row(s) affected
5	12:37:54	DESCRIBE Books	6 row(s) returned
6	12:38:33	INSERT INTO Books (title, author, genre, description, published_year) V...	10 row(s) affected Records: 10 Duplicates: 0 Warnin...
7	12:39:51	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned
8	12:40:36	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings...

123DESCRIBE Books;

124

125UPDATE Books

126SET embedding = JSON_ARRAY(0.00, 0.80, 0.20);

100%16:123

Result Grid

Filter Rows: Search

Export:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Result 4

Action Output

	Time	Action	Response
4	12:36:55	CREATE TABLE Books(book_id INT AUTO_INCREMENT PRIMARY KEY, title VAR...	0 row(s) affected
5	12:37:54	DESCRIBE Books	6 row(s) returned
6	12:38:33	INSERT INTO Books (title, author, genre, description, published_year) V...	10 row(s) affected Records: 10
7	12:39:51	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned
8	12:40:36	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0
9	12:41:13	DESCRIBE Books	7 row(s) returned

PART 5 — Add the mock vectors

159 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)

160 WHERE book_id = 9;

161

162

163 • UPDATE Books

164 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)

165 WHERE book_id = 10;

166

167

168 • SELECT

169 book_id,

170 title,

171 embedding

172 FROM Books

100% 18:173

Result Grid

Filter Rows: Search

Edit: Export/Import:

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
HULL	HULL	HULL

Books 7

Action Output

	Time	Action
✓ 19	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 20	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7
✓ 21	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8
✓ 22	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9
✓ 23	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10
✓ 24	13:37:41	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1...

PART 6 — L2 Distance Search

```
192 AS DECIMAL(10,6)
193 ) - 0.80,
194 2
195 )
196 +
197 POW(
198 CAST(
199 JSON_EXTRACT(embedding, '$[2]')
200 AS DECIMAL(10,6)
201 ) - 0.20,
202 2
203 )
204 ),
205 4
206 ) AS l2_distance
207 FROM Books
208 WHERE embedding IS NOT NULL
209 ORDER BY l2_distance ASC
210 LIMIT 3;
```

100% 9:210

Result Grid Filter Rows: Search Export: Fetch rows:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Result 10

Action Output

	Time	Action	Response
✓ 22	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Row
✓ 23	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Row
✓ 24	13:37:41	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned
✓ 25	13:38:17	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned
✓ 26	13:48:22	SELECT book_id, title, author, genre, ROUND((CAST(...	3 row(s) returned
✓ 27	13:49:13	SELECT book_id, title, author, genre, ROUND(SQRT(POW(...	3 row(s) returned

PART 7 — Cosine Similarity

```
271 ) AS cosine_similarity
272 FROM Books
273 WHERE embedding IS NOT NULL
274 ORDER BY cosine_similarity DESC
275 LIMIT 3;
```

100% 1:280

Result Grid Filter Rows: Search Export: Fetch rows:

	book_id	title	author	genre	cosine_similar...
	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 9

Action Output

		Time	Action
✓	21	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8
✓	22	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9
✓	23	13:37:41	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10
✓	24	13:37:41	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1...
✓	25	13:38:17	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1...
✓	26	13:48:22	SELECT book_id, title, author, genre, ROUND((CAST(...